

Project 65: Voice Cloning Mini Project

Goal: Clone a speaker's voice using a **few seconds of audio** and synthesize custom speech in that voice.

Techniques Used

- **Speaker Embedding:** Extract speaker identity (from audio)
 - **Text-to-Speech:** Generate speech from text using that voice
 - **Frameworks:**
 -  [Coqui TTS](#) (open-source)
 -  [Real-Time Voice Cloning \(RTVC\)](#)
 - [ **Resemble AI / ElevenLabs**] (cloud, for realism)
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Step-by-Step with Coqui TTS (Offline Approach)

Step 1: Install Dependencies

```
pip install TTS
```

Step 2: Download a Pretrained Voice Cloning Model

```
tts --list_models | grep speaker-embedding
```

Pick a multi-speaker model, like:

```
tts --model_name tts_models/multilingual/multi-dataset/your_tts
```

Step 3: Prepare Voice Sample

- Use a `.wav` file of someone speaking (≥ 3 seconds recommended)
- Make sure it's **mono, 16kHz, WAV format**

```
sample_path = "speaker_sample.wav"
```

Step 4: Clone Voice from Sample

```
from TTS.api import TTS

# Load pretrained cloning model
tts = TTS(model_name="tts_models/multilingual/multi-dataset/your_tts")

# Extract speaker embedding
speaker_wav = sample_path
text = "Hello, this is your cloned voice speaking!"

# Generate speech using the cloned voice
tts.tts_to_file(text=text, speaker_wav=speaker_wav, file_path="cloned_voice.wav")
```

Output

- You'll get a WAV file (`cloned_voice.wav`) that says your input text in the cloned voice!

Key Concepts

- **Speaker Embedding:** Captures vocal tone, pitch, and timbre
 - **Voice Transfer:** Applies that voice style to arbitrary text
 - **Few-shot cloning:** Works with as little as 3–5 seconds
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Project Summary

- **Input:** Short voice sample + target text
- **Output:** Synthesized audio mimicking the original speaker
- **Framework:** Coqui TTS (`your_tts` or `vits` + speaker embedding)
- **Use Cases:** Personalized AI, content creation, accessibility